



TECHNISCHE UNIVERSITÄT BERLIN
Faculty VII Economics and Management
Institute for Technology and Management
Chair for Entrepreneurship and Innovation Management

T08 - GreenBond
Connecting was never that easy

Learning Reflection
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Handed in to Aamruth Shankar, MSc Charleen von Kolpinski, Dr. Karina Cagarman
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1. Minimum Viable Product

1.1 Introduction

It took our team a long time to figure out what we wanted to work on. We thought about digital devices tracking the mental state of autistic people, developing biodegradable urns, or creating an alternative light-source from fluorescent fungi.

To unite our expertise and backgrounds, we decided on working on a technical device that has a positive impact on the environment and sustainability. We came up with GreenBond: an app that many of us would have needed at some point of our lives. The mission statement of GreenBond is: "We connect people to each other and their environment, promoting health, happiness, and thriving communities through nature-based activities and education."

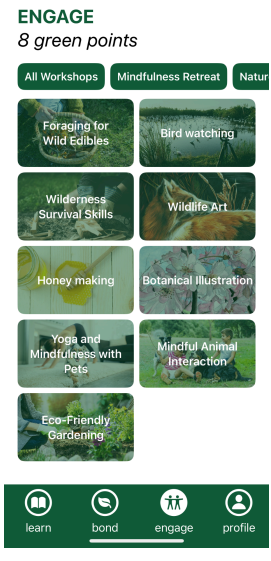
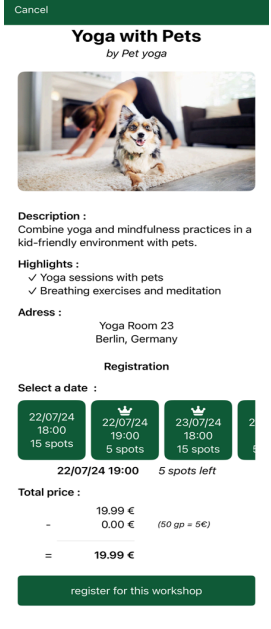
1.2 Product Details

The product consists of a mobile application that has three main themes or features: learn, bond, and engage. All three of them are explained more below, with the help of the customer Journey of our persona Kevin.

As an international student in Berlin, and like many others, Kevin often feels sad and lonely. Living in a bustling city far from home, friends, and familiar surroundings, he finds himself isolated. Unfortunately, his story is not unique. In Germany, 24% of young adults aged 18-39 experience chronic loneliness (Statista, 2024), a significant increase from 28% pre-COVID (Huber and Berlin, 2024). Recognizing that loneliness can be as deadly as smoking 15 cigarettes a day (Seitz, 2023), he looked for a solution.

GreenBond leverages the healing power of nature to combat loneliness. Through various activities and engagements, Kevin learned how spending time outdoors can significantly improve his well-being. By participating in challenges, engaging in workshops about the local ecosystem, he connected deeply with both his surroundings and himself, and met like-minded individuals. Let's explore how GreenBond works to make connecting easier and loneliness a thing of the past.

LEARN Practice mindfulness by paying close attention to the sights, sounds, and smells of nature. This can enhance the therapeutic effects. Aim to spend at least 20-30 minutes walking in a natural setting daily. This can significantly reduce stress and improve overall well-being. Regular exposure to nature boosts immune function by increasing natural killer cell activity and anti-cancer proteins. Exposure to natural environments is linked to increased serotonin levels, which can improve mood and reduce symptoms of depression.    	Feature Name: Learn Path: Main -> Learn Description: As a user, Kevin wants to access informative content about loneliness and nature. He can browse through articles, videos, and infographics to learn about the positive impact of nature on mental health and strategies to combat loneliness. This view serves as an educational resource to inform and inspire Kevin as well as all other users.
BOND July 34% Spend 15 minutes sitting quietly in nature, listening to the sounds around you. Identify at least three different sounds (e.g., birds, wind, water). --/--/3 gp Take a book or listen to a podcast while sitting in a park. Spend at least 30 minutes enjoying the content and the surroundings. --/--/3 gp Watch the sunrise or sunset from a scenic outdoor location. Take a picture and reflect on the experience. 13/07/24 4 gp Find and photograph three different types of flowers. Research and note down their names and a fact about each. --/--/6 gp    	Feature Name: Bond Path: Main -> Bond Description: Each month, Kevin can participate in a set of different challenges in nature. It is scientifically proven that nature can significantly reduce feelings of loneliness. Kevin can complete challenges such as hiking, bird watching, or tree planting to improve his mental well-being. In that page, Kevin can see his monthly progression as well as the challenges done.
Cancel Save Done Take a book or listen to a podcast while sitting in a park. Spend at least 30 minutes enjoying the content and the surroundings.  Do you feel alone today? Yes <input type="range"/> No Strongly Yes Description:	Feature Name: Comment Path: Main -> Bond -> Challenge N Description: After selecting a challenge, Kevin can document his experience with photos or comments. He can also monitor his feelings to track and improve his overall well-being. By completing challenges, he will earn Green Points which he will be able to use in the Engage mode.

	Feature Name: Engage Path: Main -> Engage Description: In the Engage mode, we provide workshops offered by our partners. As a user, Kevin can book these workshops or attend social events and connect with others who have shared interests or hobbies. These workshops or social events aim at enhancing Kevin's social engagement and cover topics like local flora and fauna, eco-friendly practices, outdoor survival skills, and many more.
	Feature Name: Workshop Path: Main -> Engage -> Workshop N Description: After selecting a workshop he wants to participate in, Kevin can use the Green Points previously earned from challenges to register for the workshop offered by partners and convert them into discounts. What's more, if he's a premium customer, he'll be able to sign up for more and a wide range of workshops.

	Feature Name: Profile Path: Main -> Profile Description: Kevin also has a personal profile where he can look back on and review his completed challenges, accumulated Green Points, and his booked workshops. This view helps him to track his progress and achievements by displaying his mental health progress and challenges.
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1.3 Limitation and Future scope

Currently, the GreenBond app is developed in Swift, which gives it stability and coding speed. However, this means that it is only available for iOS devices and furthermore, limits accessibility for users of other platforms such as Android, potentially excluding a significant proportion of the target audience. Broadening the app's availability by developing versions for Android and web platforms will increase accessibility and enable a wider audience to benefit from the features.

In addition, the application relies on a trust system to verify whether users have completed the challenges. This could lead to inaccurate reporting and reduced challenge-based engagement effectiveness if users do not honestly report their activities. Integrating artificial intelligence, such as computer vision for image recognition, to check completion of challenges will ensure that users have actually committed to the activities. Also, AI could provide personalized feedback and suggestions based on user performance and preferences.

Moreover, as our app is based on three functions, learning, bonding, and engaging, the app will need to be updated regularly, including new articles and diversified challenges to maintain user interest. Additionally, partnerships with (nature-related) local organizations and mental health professionals would also need to be expanded and maintained to offer a wider range of workshops and activities to the customer including Kevin.

Finally, the future plan would be to also launch the app in the United States, due to the steady growth of mental health apps and relatively low stigma around mental health. However, this would need considerable investments that should be taken into account. For example, the app should be translated and working in English and new partners should be looked for. We will need the help of investors to support the extension of GreenBond, for which a successful launch in Germany would be needed in order to prove the viability of the concept.

2. Project Management

Participating in this team project was a valuable learning experience that allowed each member to contribute their unique skills and perspectives toward a common goal. Reflecting on our journey, it becomes clear that our collaboration and mutual support were key to our success.

2.1 Individual Team Member Roles

Zied Baatour was the one who initially came up with the idea of tackling loneliness, which formed the foundation of our project. He focused on writing the problem, i.e., the business idea section, outlining the significance and urgency of addressing loneliness, discussing the idea's value proposition in our business plan, the degree of novelty of our product, and the unique selling point, as well as outlining the revenue model. Additionally, he contributed to the executive summary, ensuring that our project's objectives and vision were clearly articulated.

Kevin Ferreira worked on the team section of our business plan, explaining our team, the members' backgrounds and roles, and personnel planning, as well as on the organization and operations, examining our business's legal structure, value chain, and internal and external activities. What is more, he developed the prototype of our app. His technical skills were crucial in bringing our concept to life and demonstrating its feasibility.

Victoria Alex worked on the sustainability impact of our product, highlighting its long-term benefits and alignment with the Sustainable Development Goals and specifically their targets and indicators. This added depth to our project, showcasing our commitment to sustainable practices. She worked on the value proposition for both our interim and final pitch, highlighting the connection between loneliness-related health issues and environmental strategies to address them. Having a great eye for detail, she also worked on the design for our slides & did final touch-ups for the presentations.

Marieke Teetz wrote the market/ industry analysis, providing insights into the market and industry we aim to enter, our target audience, and current market trends/ drivers that add potential to our idea. She worked on both market sizing and competitor analysis. Additionally, she wrote the initial outlines for both our LinkedIn posts as well as the financial part of the business plan.

Tarek Ramadan's expertise in marketing was essential for developing strategies to promote our product. He worked on segmentation, targeting, and positioning, how to develop what kind of customer relationship, pricing mechanisms, and our general advertising and visibility strategy. His work ensured that our marketing approach would be effective.

Evelien Pigeaud wrote the business strategy section. She worked on both the business assessment and risk management as well as the market entry strategy, determined milestones for our business, and worked on future scenarios and developments. For overall flow and coherence of the report, she has critically read the report to take out mistakes, improve readability and structure. Lastly, she has written the executive summary, with the help of corrections from Zied.

Collaborative Efforts:

Together as a team, we worked on the financials. During Bootcamp Day, we spend the day working on the financial sheet and determining pricing strategies and mechanisms. We also all worked on the pitches, each working on the slides that cover their sections in the business plan. We have had meetings with all of us to go over the pitches, make sure we all agreed on the content and layout. Moreover, we practiced while we all presented for the interim pitch, only three of us, namely Tarek, Victoria, and Kevin, presented during the final pitch. Marieke, Zied, and Evelien prepared for the question part.

Our almost weekly meetings, both with our mentor Maik Mathey and within the team, provided regular opportunities to discuss progress, address challenges, and refine our approach. Maik has also given some valuable tips on improving the business plan. Moreover, when working on the plan, we made sure to critically read the parts of others, comment and suggest improvements.

Communication was a strong point in our collaboration. We fostered an environment where everyone felt comfortable discussing problems and seeking solutions collectively. This openness helped us to navigate any issues that arose efficiently.

2.2 Issues & overcoming them

Our team was composed of members from different academic backgrounds, which often led to diverse ideas and approaches for our project. While this diversity enriched our

project with varied perspectives, it also posed challenges. At times, these differing viewpoints made it difficult to work together seamlessly. For example, it took us some time to agree on the same topic for the project, as well as on the main features and focus of the app. However, we always found a way to resolve arguments or disagreements constructively. By actively listening to each other and seeking common ground, we were able to integrate our diverse insights into a cohesive and innovative project.

Despite the occasional difficulty in meeting internal deadlines, we always managed to submit everything on time. We strategically set early deadlines to allow for any delays, demonstrating our proactive approach to project management.

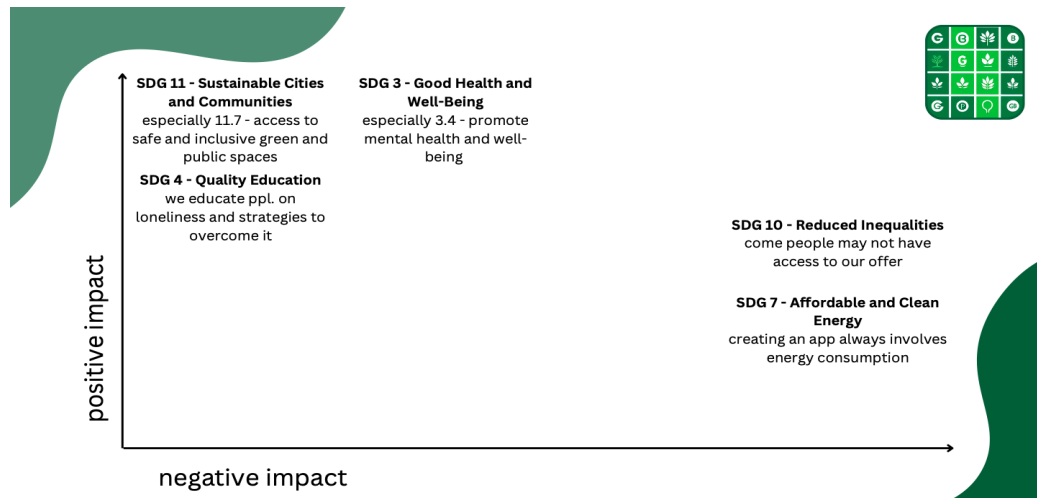
2.3 Project Betterment

While we were a great team and worked together effectively, we had issues settling on an idea, which made the ideation last a long time. We would have been able to start sooner into diving deep into our business idea if we would have settled on an idea sooner. This would possibly have improved some of the weaker points that were identified about the app during the pitch. Additionally, we could have followed the Design Thinking steps more precisely and accurately as it would maybe have given an additional structure to our work.

Overall, we showed great teamwork, effective communication, and the ability to navigate diverse perspectives. Each member's dedication and expertise were integral to our success, and our collective efforts resulted in an impactful project that we are all very passionate about. This experience has reinforced the importance of collaboration, adaptability, and proactive planning in achieving project goals. By embracing our differences and leveraging our strengths, we were able to overcome challenges and deliver a project we are all proud of.

3. Sustainability Impact

3.1 Initial Impact



At the beginning of the GreenBond project, the sustainability impact forecast focused on three primary Sustainable Development Goals (SDGs): SDG 3 (Good Health and Well-Being), SDG 4 (Quality Education), and SDG 11 (Sustainable Cities and Communities).

GreenBond aims to significantly enhance mental health by reducing loneliness through nature-based activities, contributing to SDG 3. By integrating educational elements on mental health and environmental awareness, GreenBond also targets SDG 4. Furthermore, by fostering community through shared nature activities, GreenBond is expected to support SDG 11 by promoting sustainable, connected communities.

However, potential drawbacks include a limited focus on reducing inequalities (SDG 10) and the environmental impact of increased e-waste and resource consumption (SDG 7).

3.2 Final Impact

After three months of business plan execution and product development, the sustainability impact of GreenBond has evolved, maintaining the initial SDG targets while expanding upon them:

SDG 3 (Good Health and Well-Being): The initial impact forecast for SDG 3 proved accurate. Research indicates significant improvements in mental well-being and reduced feelings of loneliness among participants in nature-based activities (Litt et al., 2024, Garside et al., 2020, Panțiru et al., 2024, Coll-Planas et al., 2024). These activities provide a therapeutic and communal environment, leading to an improved mental health.

SDG 4 (Quality Education): In the Learn mode of GreenBond, users will be engaging with the educational modules, which increase awareness and knowledge about the

relationship between nature, mental health, and environmental issues. That is why the forecast proved to be accurate.

SDG 11 (Sustainable Cities and Communities): GreenBond will increase collaborations with local organizations to promote sustainable urban development. Initiatives like community gardening projects and reducing loneliness among the communities can further solidify GreenBond's role in promoting sustainable communities.

SDG 10 (Reduced Inequalities): Initially, we believed that we may impact this SDG negatively as not all people have access to phones and, hence, access to our offer. However, more than 60 million people in Germany own a smartphone, which is more than we expected. We will be working on developing the app for all (iOS and Android) devices. To enhance accessibility, GreenBond will be available as a freemium model, meaning that users can but do not have to purchase the premium version. Additionally, we are committed to including all age groups by developing user-friendly tutorials and support for elderly users to reduce digital inequalities.

SDG 7 (Affordable and Clean Energy): Initially, we believed that, as app developers, we will have a negative impact on the environment as digital development always includes energy consumption or may increase e-waste. However, we found out that developing the app and keeping it running does not include as much power and energy as we initially thought it would. Hence, we believe that this SDG will not be as negatively impacted by us as assumed. We will also focus on using sustainable cloud providers to reduce energy consumption. For this reason, in the future we would like to engage ourselves with SDG 12 (see 3.3).

3.3 Future Impact

SDG 12 (Responsible Consumption and Production): In the future, GreenBond will prioritize using eco-friendly, sustainable cloud providers to reduce electricity consumption. This initiative aims to minimize the environmental footprint associated with digital operations. By adopting green cloud solutions, we will contribute to a more sustainable digital ecosystem. Moreover, with nature as one of the focal points of the application, customers may feel more inclined to also behave more responsibly towards it.

GreenBond will have an overall positive impact on the environment and people's mental health. It will also positively impact the German health care system since people will be less lonely and more happy, meaning they will need less (financial) support.

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